energies

MDPI

Article

A Unified Online Assessment Framework for Pre-Fault and Post-Fault Dynamic Security

Xin Li [1,2], Rongkun Shang [1], Qiao Zhao [3,4,*], Yaowei Zhang [1], Jingru Liu [1], Changjie Wu [1] and Panfeng Guo [1,2]

[1] College of Electrical Engineering and New Energy, China Three Gorges University, Yichang 443002, China; ctgulx@ctgu.edu.cn (X.L.); ctgusrk@ctgu.edu.cn (R.S.); ctguzyw@ctgu.edu.cn (Y.Z.); ctguljr@ctgu.edu.cn (J.L.); wuchangjie@ctgu.edu.cn (C.W.); ctgugpf@ctgu.edu.cn (P.G.)
[2] Hubei Provincial Engineering Research Center of Intelligent Energy Technology, China Three Gorges University, Yichang 443002, China
[3] Power China Guiyang Engineering Corporation Limited, Guiyang 550000, China
[4] College of Hydraulic and Environmental Engineering, China Three Gorges University, Yichang 443002, China
* Correspondence: zhaoqiao_gyy@powerchina.cn

Abstract

With the expansion of interconnection in power systems and the extensive adoption of phasor measurement units (PMUs), the secure operation of power systems has been increasingly covered in research. In this article, a unified online framework for pre-fault and post-fault dynamic security assessment (DSA) is proposed. First, maximum mutual information (MIC) and the random subspace method (RSM) are employed to select the key variables and enhance the diversity of input data, serving as feature engineering. Then, a deep forest (DF) regressor and classifier are utilized respectively to predict security margin (SM) and security state (SS) during online pre-fault and post-fault DSA based on the selected variables. In pre-fault DSA, scenarios with high SM are identified as stable, while those with low SM are forwarded to post-fault DSA. In addition, a time self-adaptive scheme is employed to balance low response time and high prediction accuracy. This approach prevents the misclassification of unstable scenarios as stable by either outputting high-credibility predictions of unstable SS or deferring decisions on SS until the end of the decision-making period. The unified framework, tested on an IEEE 39-bus system and a practical 1648-bus system provided by the PSS/E version 35 software, demonstrates significantly improved assessment accuracy and response times. Specifically, it achieves an average response time (ART) of 2.66 cycles for the IEEE 39-bus system and 3.13 cycles for the 1648-bus system while maintaining an accuracy exceeding 98%, surpassing the performance of currently widely used deep learning models.

Keywords: feature engineering; deep forest; dynamic security assessment; time self-adaptive scheme

Received: 31 December 2025
Revised: 25 January 2026
Accepted: 26 January 2026
Published: 27 January 2026

Copyright: © 2026 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

As the scale of interconnection in power systems expands and network loads escalate, the operational complexity and uncertainty of these systems increase significantly [1]. This development trend heightens the possibility of power systems operating near their security boundaries [2,3]. Disturbances like load fluctuations, generation failures, or the loss of critical elements can challenge system stability, making the risk of breaching

security constraints more pronounced. In this context, dynamic security assessment (DSA) for operation points (OPs) emerges as a crucial process, evaluating a power system's response and resilience to these disruptions [4].

Each OP contains a wealth of security information [5]. However, such information is highly dimensional, complex and hard to analyze. Traditional DSA method time domain simulation, extended equal area criteria, and transient energy functions [6–8] cannot evaluate OP security with both speed and accuracy. The advent of Wide Area Measurement Systems (WAMSs) and the adoption of phasor measurement units (PMUs) [9] facilitate the development of data-driven DSA approaches. These approaches expedite the computational process while ensuring accuracy and reliability by establishing mappings between system variables and the security margin (SM)/security state (SS) through offline training. This establishment of mapping relations enables swift online assessment using real-time PMU data from WAMSs. In [10], a novel continuous-label transient rotor angle SM is used to measure the distance of the current OP from the security boundary. In [11], diverse cases of time series short-term voltage SM are divided by a u-shapelet-based hierarchical clustering data-driven method. In [12], a gated recurrent unit (GRU)-based model is proposed to predict the SS of the system after faults lasting 0.25 s using the current OP information. In [13], real-time static voltage SS on current OP is assessed by an integrated data-driven scheme, based on an efficient Neyman–Pearson Umbrella (NPU) algorithm. In [14], an improved CNN model for DSA is proposed to fully mine the temporal information of the measurement data in the process of a power system. In [15], a DSA model based on a Transformer encoder is proposed. In [16], a data-driven approach that integrates Graph Convolutional Networks (GCNs) with Long Short-Term Memory (LSTM) Networks is proposed to analyze the stability of transient power angles. In [17], a physics-augmented auxiliary learning (PA-AL) framework for TSA is proposed, implemented using multiple mixtures of expert models within a physics-informed paradigm for end-to-end stability margin prediction. In recent years, deep learning techniques have gained significant prominence in the research of DSA. However, these approaches present significant challenges. They typically require large amounts of training data and extensive hyperparameter tuning to prevent overfitting. Additionally, their complex network architectures often lead to high computational costs during offline training and increased latency during online inference, which can be detrimental to real-time applications.

Data-driven DSA methods can be categorized into two phases based on the timing of the assessment: pre-fault and post-fault [18]. Current research in data-driven DSA primarily focuses on the post-fault phase [12–20]. The former utilizes steady-state data before the occurrence of faults to calculate the distance between the current OP and the security boundary, aiming to proactively identify and mitigate potential uncertainties. Given the complexity of power systems and the low probability of instability, this preemptive assessment is essential. In this context, ref. [21] proposes a multi-label learning-based DSA method that is capable of performing stability assessment for multiple different fault scenarios before the fault occurs. On the other hand, the latter focuses on the dynamic data after a fault occurs, monitoring the system behavior post-fault removal to swiftly implement stabilization measures if instability signs appear. Most studies address these two phases separately, which highlights a key limitation. There is a temporal correlation between the pre-fault and the post-fault, and the disjoint analysis of pre-fault and post-fault scenarios overlooks the role of pre-fault state identification as prior knowledge that aids in the training of post-fault scenarios. If pre-fault and post-fault data can be combined, and a comprehensive analysis of the time series characteristics over the entire period is conducted, the accuracy of the safety assessment can be significantly improved. Some studies have attempted to combine them; for example, ref. [22] combines pre-fault and post-fault information by extracting features using Graph Neural Networks (GNNs) and

by predicting dynamic trajectories with Fourier Neural Operators (FNOs). However, most studies have not established a unified framework to integrate the pre-fault and post-fault assessment tasks. Therefore, integrating pre-fault and post-fault information into a unified framework remains a challenge.

Although data-driven DSA methods can effectively construct mapping relations, the vast number of input features involved do not all positively impact the final prediction [19]. Redundant information can lead to prolonged training times and decreased prediction accuracy. Feature engineering is a pivotal approach to addressing this issue, involving the process of identifying features that more accurately describe the underlying problem. The two principal techniques within feature engineering are feature selection and feature extraction. Feature selection algorithms [23] are primarily used to identify the most relevant features and eliminate redundant ones, while feature extraction [24] involves mapping input vectors to a new vector space, thereby generating new features that significantly contribute to the predictive results. These techniques enhance the efficiency and accuracy of the model by optimizing the input feature set.

To address these limitations, this paper proposes a unified online DSA framework based on deep forest (DF) [25]. This framework combines pre-fault screening with post-fault evaluation. It identifies high-SM stable conditions in the pre-fault stage and transmits this information to the post-fault assessment. This integration of pre-fault and post-fault information effectively enhances computational efficiency and assessment accuracy. Unlike deep neural networks, the DF model is capable of handling high-dimensional data with fewer hyperparameters and lower computational costs, making it more suitable for practical engineering applications. To construct an efficient DSA framework suitable for practical implementations, the maximal information coefficient (MIC) [26] and random subspace method (RSM) serve as feature engineering tools, and an advanced DF regressor and classifier are applied respectively to predict SM and SS. The MIC is particularly chosen for its superior ability to capture both complex linear and nonlinear relationships in large datasets. The RSM using multi-grained scanning plays a major role in enhancing the diversity of selected features by inputting various windows for data extraction. Furthermore, the DF approach, integrating boosting algorithms with a cascade forest structure, is employed for regression and classification tasks, specifically targeting continuous label SM and discrete label SS. Additionally, the RSM and the DF exhibit inherent compatibility in algorithm design, which significantly reduces the complexity of model integration [24]. During the online stage, data-driven methods for post-fault DSA might encounter challenges with decision-making speed due to the preset response time, ranging from four cycles to 3 s [27,28]. Despite this, achieving short response times typically necessitates a compromise in accuracy. Consequently, a time-adaptive evaluation method is applied to manage this trade-off effectively.

Based on tests for the IEEE 39-bus system and a practical 1648-bus system, the feasibility and efficiency of the proposed unified DSA framework are demonstrated. From a practical point of view, the main contributions of this paper can be summarized as follows:

1. A unified pre-fault and post-fault framework is proposed for online DSA, containing three stages: data acquisition and feature design, offline training, and online assessment.
2. A new feature engineering approach is proposed to manage redundant inputs, demonstrating improved data-driven model performance.
3. A time self-adaptive scheme is proposed to improve post-fault DSA performance, leading to greater accuracy and significantly reduced response times.

The rest of this paper is organized as follows: Section 2 includes the problem statement and introduction of the supporting methods. Section 3 introduces the whole application process of the DSA framework. Section 4 presents the performance test of the

proposed framework in the IEEE 39-bus system and a practical 1648-bus system. Finally, the paper is concluded in Section 5, where a summary of the findings is given, and directions for future studies on the topic are provided.

2. Problem Description and Methodology Statement

2.1. SM/ST of Pre-Fault/Post-Fault Security Assessment

The security of a steady-state power system is defined by its ability to withstand imminent disturbances while maintaining a continuous power supply to end users. In the pre-fault DSA, SM is utilized to quantify the distance between the current OP and the security boundary. The critical clearing time (CCT) refers to the longest duration for which the power system can tolerate the presence of a fault [10]. The primary objective of the pre-fault DSA in our framework is to select absolutely stable samples. Therefore, the pre-fault DSA in this paper adopts a conservative approach, with consideration given to the performance of the regressor. To evaluate the security margin of an OP under various fault conditions, all CCT values for different fault scenarios at that operating point are considered. As shown by Equation (1), the value of the minimum CCT ($CCT_{i(\min)}$) among different faults is adopted to represent the SM of the OP. It is particularly noted that samples with an SM greater than the threshold are absolutely stable. By fixing the SM at the worst-case baseline, this paper ensures the selection of absolutely stable samples, thereby reducing the probability of misclassifying unstable samples as stable. This approach effectively alleviates the computational burden of subsequent fault assessments.

$$SM_i = CCT_{i(\min)} \tag{1}$$

where SM_i represents the $CCT_{i(\min)}$ at OP_i.

When the system is subjected to a fault, the security refers to the ability of the power system to maintain or restore an acceptable state of operating equilibrium. In post-fault DSA, the criterion for determining SS [12–17], as given by Equation (2), is whether the maximum rotor angle difference between any two generators exceeds 360° in the 3–6 s after the fault is cleared. It is important to note that Formula (2) defines a criterion for transient stability, which is utilized to derive the true labels during the offline training phase. During the offline model training phase, the entire transient process is typically used as an input to accurately capture both fast and slow instability phenomena. In contrast, for the online application phase, the post-fault DSA is designed to predict this future stability outcome using only early post-fault data. This predictive approach is essential to provide a sufficient time margin for emergency control before physical instability actually occurs.

$$\eta = \frac{360^\circ - |\Delta\delta|_{\max}}{360^\circ + |\Delta\delta|_{\max}} \tag{2}$$

where $|\Delta\delta|_{\max}$ is the maximum rotor angle difference between the outputs of any two generators. When η is greater than 0, the sample is judged to be stable; otherwise, it is judged to be unstable.

2.2. MIC

The MIC is employed to select features with strong correlation and low redundancy in this paper. According to the definition given by Reshef et al. [26], given random $A = \{a_i, i = 1, 2, \cdots, m\}$ and $B = \{b_i, i = 1, 2, \cdots, m\}$, where m is the total number of samples, the mutual information between A and B can be defined as

$$MI(A,B)=\sum_{a\in A}\sum_{b\in B}p(\mathrm{a},\mathrm{b})\log\frac{p(\mathrm{a},\mathrm{b})}{p(\mathrm{a})p(\mathrm{b})} \tag{3}$$

where $p(\mathrm{a},\mathrm{b})$ is the joint probability density function of A and B. $p(a)$ and $p(b)$ are the marginal probability densities of A and B, respectively. For a given ordered pair, (A,B), the dataset is denoted as $D=\{(a_i,b_i)\}\in R^2$ with a sample size of m. The dataset D is partitioned into a grid, G, with x columns and y rows. The mutual information, $MI(A,B)$, is calculated for each grid cell formed by this partitioning. Since there are multiple ways to partition the data into $x*y$ grids, the maximum mutual information value, $MI*$, across these different partitioning schemes is taken as the mutual information value for grid G.

$$MI^*(D,x,y)=\max MI(D\mid G)) \tag{4}$$

Subsequently, $MI^*(D,x,y)$ is normalized by $\log\min\{x,y\}$ as follows:

$$M(D)_{x,y}=\frac{MI^*(D,x,y)}{\log\min\{x,y\}} \tag{5}$$

The value of $I^*(D,x,y)$ falls from 0 to 1 in Formula (5). Then, the value of the MIC for two variables, A and B, is given as follows:

$$MIC(D)=\max_{xy<B(n)}\{M(D)_{x,y}\} \tag{6}$$

where $xy<B(n)\{B(n)=n^{\alpha}\}$ is the limit for the grid size.

The interval of the MIC is $[0,1]$, and the higher MIC value indicates a stronger correlation between the investigated features.

2.3. RSM

The proposed method for extracting influential features in prediction outcomes involves a novel random subspace method based on multi-grained scanning [22]. To illustrate the approach, consider an initial set of 40 raw features, as depicted in Figure 1. A 10-dimensional sliding window, highlighted in red, scans the input data to generate a feature vector by sliding across each feature, yielding 31 vectors. Subsequently, 15 vectors are randomly selected through subsampling from both positive and negative examples to serve as training instances. These instances are then used to train both a random forest and a completely random tree forest classifier for binary classification, each of which produces a 2-dimensional class vector, resulting in 30 class-distribution features. The RSM then aggregates these into a 60-dimensional feature vector for representation.

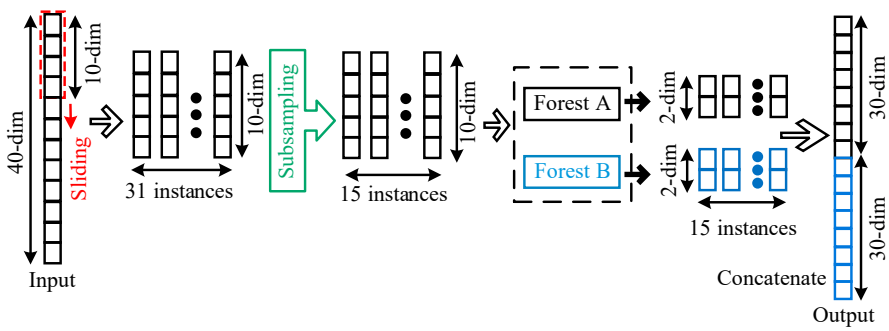

Figure 1. Illustration of random subspace method based on a sliding window scan.

Similarly, for regression, new instances are used to train both a random forest and a completely random tree forest regressor. The regressor outputs prediction values for the SM, which are then concatenated into a representation feature vector. This process ensures a coherent and efficient feature extraction and representation for both classification and regression tasks. It is worth noting that the sliding window mechanism primarily functions as a feature enhancement strategy. By scanning local subsets of the feature vector, RSM generates a set of diversified transformed features. Although different feature orderings can alter the specific combinations within each window, the subsequent cascade forest stage effectively integrates these enhanced features to capture the global decision boundary, encouraging the model to learn interactions between different variable groups.

To further mine useful information from original features, N windows of varying sizes can be used to scan raw features. The differently grained feature vectors will be generated, as shown in Figure 2.

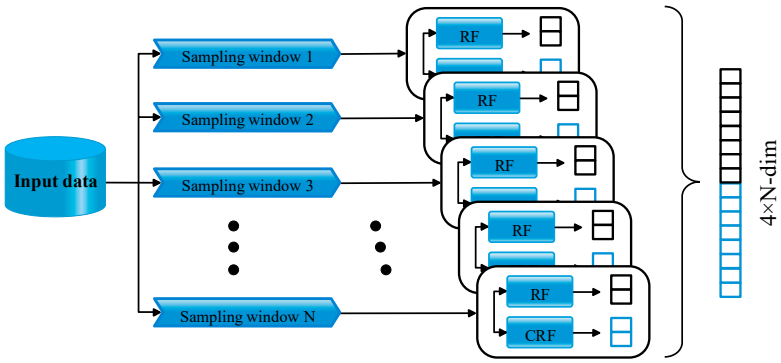

Figure 2. Illustration of random subspace method using multi-grained scanning.

2.4. DF

The DF, a predictive method applicable to both classification and regression scenarios, utilizes a cascade structure depicted in Figure 3. At each cascade level, the input consists of the initial vector and the output vector from the previous level. For the classification tasks, the output of each level is a series of class vectors. The final prediction is obtained by aggregating the class vectors of the last level.

Each level of the decision forest (DF) is an integration of multiple forests, and each forest is composed of decision trees. Thus, the DF represents an ensemble approach of decision trees. Let $(x_1, x_2, \ldots, x_m, Y, U)$ be a dataset, where x_i has domain $D(x_i)$, Y has domain $D(Y) = \{1, 2, \ldots, J\}$, and U is the random variable. The function $\psi : D(x_1) \times \cdots \times D(x_m) \rightarrow D(Y)$ can be constructed by the decision tree classifier. An illustration of the classification process for a decision tree classifier is shown in Figure 4. Based on input features, $(x_1, x_2, \ldots, x_5)$, the trained samples are recursively divided into hypothetical categories at each nonterminal node to establish splitting rules of the nodes. For a sample, it is passed down along the decision tree path, starting from the root node until it reaches a terminal node. For a fully grown decision tree, each terminal node represents a category.

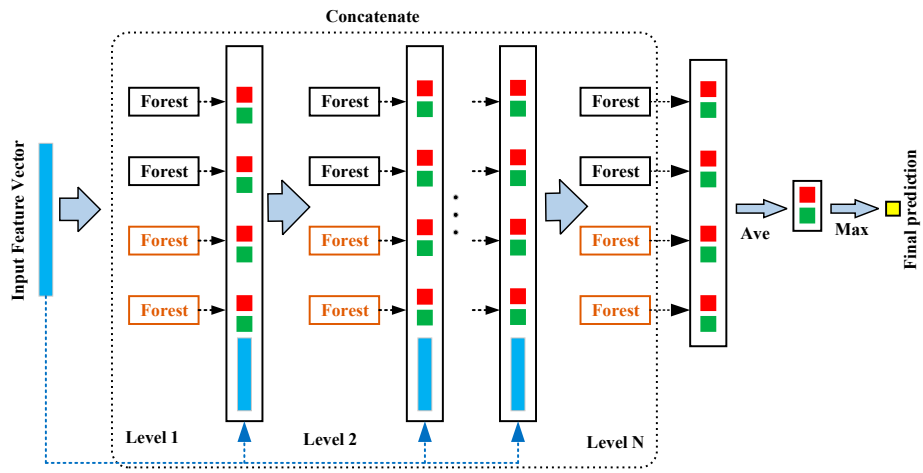

Figure 3. Illustration of the deep forest structure.

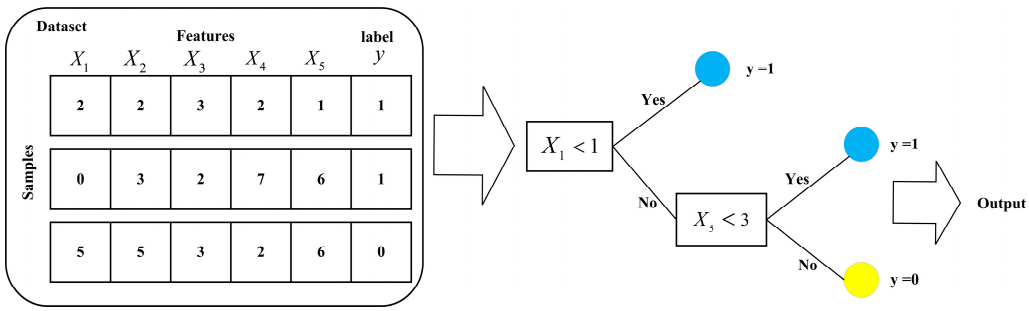

Figure 4. Illustration of classification process of a decision tree.

A critical step in constructing a decision tree is selecting an appropriate feature and its corresponding split value at each nonterminal node. The information gain ratio and Gini index are two indices used to measure the growth of data purity after splitting. Higher purity means better splitting.

Similarly, the function $\varphi : D(x_1) \times \cdots \times D(x_m) \to U$ can be constructed by the decision tree regressor. The construction process is as follows:

Supposing a splitting feature, x_j, and a splitting point, x_s, two domains are defined as $R_1(j,s) = \{X \mid X^{(j)} \le X_s\}$ and $R_2(j,s) = \{X \mid X^{(j)} > X_s\}$. The best x_j and x_s can be searched by minimizing the objective function (7):

$$\min_{j,s}[\min_{c_1} \sum_{x_i \in R_1(j,s)} (y_i - c_1)^2 + \min_{c_2} \sum_{x_i \in R_2(j,s)} (y_i - c_2)^2] \tag{7}$$

where c_1 and c_2 are output values of two domains, which can be calculated by Equation (8).

$$c_i = ave(U_i \mid X_i \in R_i) \tag{8}$$

The decision tree regressor can be generated by constructing decision rules at each nonterminal node. The output of the terminal node can be calculated by Equation (8).

Based on different rules, the decision trees can be integrated into the decision tree forest. The most commonly used decision tree forests include random forests [12] and completely random tree forests. A random forest is generated by randomly selecting $\sqrt{m}$ number of critical features as candidates and choosing the one with the best purity gain for splitting at each node (m is the number of input features). In contrast, a completely random tree forest is generated by randomly selecting a feature for splitting at each node.

The output of the decision tree forest is obtained by averaging the outputs of the individual decision trees.

3. Proposed Approach for Pre-Fault and Post-Fault DSA

A unified approach for pre-fault and post-fault DSA is introduced in this study, including feature engineering and a deep forest model. The proposed method for online security assessment consists of three stages: (1) data acquisition and feature design, (2) offline training, and (3) online assessment. The detailed information of the three stages is shown in Figure 5.

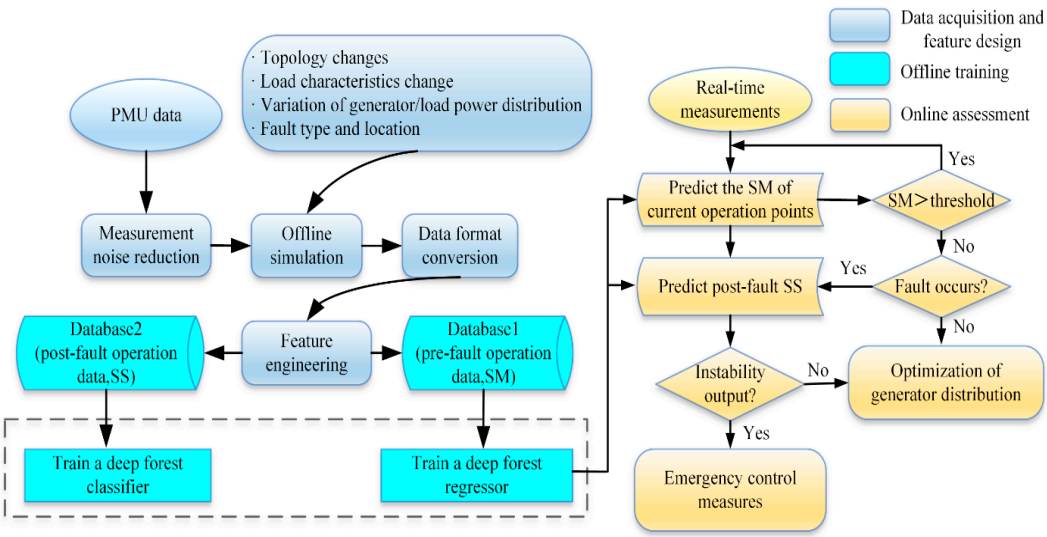

Figure 5. The proposed unified prediction scheme for pre-fault and post-fault security.

3.1. Data Acquisition and Feature Design

An effective data acquisition system is crucial for the data-driven method. To meet the dual requirements of collecting diverse operation conditions and swiftly monitoring system instability, PMUs and WAMS are well-suited for the proposed pre-fault and post-fault DSA. While historical PMU data provide a rich source of training samples, gaps in operational conditions necessitate the use of time-domain simulations and linear interpolation for a more comprehensive dataset, including scenarios like topology changes and equipment outages. This enriched dataset serves as the foundation for the pre-fault and post-fault DSA.

The system variables sampled at PMU installation locations serve as direct features to reflect the security of the power system. The raw feature set can be constructed using either individual variables or combinations of them. In this study, we select bus voltage amplitude; phase angle; active and reactive power on both the generator and the transmission line; and the generator rotor angle as the raw features. To accommodate the distinct physical characteristics of the unified framework, the input data is structured differently for the two assessment stages. The pre-fault assessment operates in a continuous monitoring mode, where the input vector at each time step represents a steady-state snapshot, processed continuously to update the security margin in real-time. In contrast, the post-fault assessment adopts a cumulative time-window approach, constructing an input matrix that captures the dynamic trajectory of system variables from the fault-clearing instant to the current decision cycle, thereby effectively capturing the temporal evolution of system instability. However, directly utilizing PMU features presents challenges due to the large volume of data and high redundancy. To address this issue, this study adopted the method of feature engineering.

The MIC algorithm first evaluates the correlation between features and labels or regression values. Features with a correlation below the threshold of 0.7 are discarded, and the top N features with the highest correlation scores are selected. Subsequently, the MIC values between the selected features are calculated, and features with redundancy exceeding the 0.7 threshold are removed to minimize overlap between features. This process ensures that the selected features retain high information content while maintaining low dimensionality. These chosen features are processed through the RSM module, which uses various scanning windows to generate re-represented features. The outputs are merged into the final dataset, which is further downsampled to reduce dimensions. This approach transforms raw features into a new vector space, optimizing the input for machine learning models.

As a result, *n* and *m* features are generated for the pre-fault and post-fault assessment models respectively. The two training databases can be expressed as (X_i, U_i) (X_j, Y_j), where $X_i = (x_1, x_2, \ldots, x_n)$; U_i is the SM of the i-th sample, $X_j = (SM, x_1, x_2, \ldots, x_{m \times l\max})$; and Y_j is the SS label of the j-th sample (e.g., +1 for stable; −1 for unstable).

3.2. Offline Training

By utilizing training databases (X_i, U_i) and (X_j, Y_j), a DF classifier and a DF regressor are trained. In each level of the DF, two random forests and two completely random tree forests are employed to enhance diversity, a crucial aspect in ensemble construction. The boosting algorithm is implemented during training. Specifically, after each new level is trained, the model's overall performance is tested. If there is no significant improvement in performance, the expansion of the cascade structure is automatically terminated.

For training dataset (X_i, U_i), the mapping relationship $\psi(X_i) = U_i$ is constructed through offline training of the DF regressor. The specific training process of the DF regressor is detailed in Algorithm 1. Training set (X_i, U_i) of the DF regressor is input, starting with the first layer, where l represents the current layer number. The training proceeds layer by layer until l reaches the maximum layer number, l_{max}. In the current layer, l, M forests are trained in sequence. The depth of each tree in the forest is D, and each forest contains n_{child} decision trees. Regarding the hyperparameter configuration of the deep forest, the number of forests, M, was set to four to ensure diverse feature representation. To mitigate the risk of overfitting and control computational costs, the tree depth, D, is limited to 10. Additionally, the decision trees, n_{child}, were set to 200. For each subsequent level l, the input vector is constructed by concatenating the original raw features with the prediction outputs from the previous level, $l-1$. This design allows the model to progressively learn deeper feature representations. Furthermore, an automatic termination mechanism is implemented to prevent overfitting. After training each level, the model evaluates the performance accuracy, q_l. If the accuracy improvement between the current layer and the previous layer is below the preset threshold, ε, the cascade expansion is automatically terminated, determining the optimal depth of the model.

Algorithm 1. Training of DF regressor.

Input: $x_1, x_2, \ldots, x_n, U$

Tuning parameters: (M, D, $l_{\max}$, n_{child})

1 For level l ($l \le l_{\max}$) do

2 Training: $m \leftarrow 1$ to M with depth D. Each forest has n_{child} decision trees.
3 Verify the prediction accuracy q_l
4 If $q_l - q_{l-1} > \varepsilon$ do
$l = l + 1$
Go to 1
5 Return DF = $\{DF_l : depth(l) = D,\ num(l) = m\}$
6 End

For training dataset (X_j, Y_j), a DF classifier is trained for post-fault security analysis. The description of the training process of the proposed prediction model is detailed in Algorithm 2. First, input dataset (X_j, Y_j) and the prediction result SM value of the DF regressor. The training begins with the first layer, where l, l_{max}, M, and n_{child} are consistent with Algorithm 1. It is worth noting that the DF classifier does not have an early stop strategy because it adopts a time-adaptive strategy and makes judgments at each level. After the training reaches the maximum number of layers, l_{max}, the training process stops, and the final DF classifier model is returned.

Algorithm 2. Training of DF classifier.
Input: $SM, X_1, X_2, \ldots, X_{l\max}, Y$, where $X_i = (x_1, x_2, \ldots, x_m)$
Tuning parameters: $(M, D, l_{\max}, n_{child})$
1 For level l $(l = 1, 2, \ldots, l_{\max})$ do
2 Input: $SM + X_l\ for\ l = 1$
3 Training: $m \leftarrow 1$ to M with depth D. Each forest has n_{child} decision trees.
4 $l = l + 1$
5 Go to 1
6 Return DF = $\{DF_l :\ level(l) = l_{\max},\ depth(l) = D,\ num(l) = m\}$
7 End

3.3. Online Assessment

The pre-fault and post-fault prediction models can be trained in the offline stage. In the application stage, the selected features are regularly sampled by PMUs. The typical captured monitor data appears as

$$\text{Data} = [X_1 \quad X_2 \quad X_3 \quad \cdots \quad X_t] \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & \cdots & x_{1,t} \\ x_{2,1} & x_{2,2} & x_{2,3} & \cdots & x_{2,t} \\ x_{3,1} & x_{3,2} & x_{3,3} & \cdots & x_{3,t} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{v,1} & x_{v,2} & x_{v,3} & \cdots & x_{v,t} \end{bmatrix} \tag{9}$$

where v is the number of features selected by the MIC.

The decision process of the proposed unified prediction framework is illustrated in Figure 6. The SM of a steady-state operation point is promptly determined when real-time PMU measurements are obtained. Assuming a fault occurs at t_f and is cleared at t_c, the most recent SM prediction value at the time of t_f is retrieved. If the SM is below the threshold, it, augmented with X_f, is input into the first level of the DF classifier to assess the transient stability of the power system.

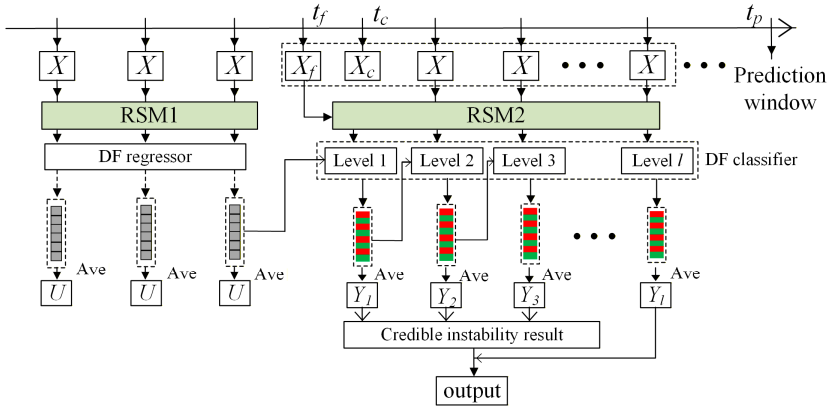

Figure 6. The decision process of the proposed unified prediction framework.

A time self-adaptive scheme is employed in online assessment, generating highly confident predictions of "stable" or "unstable." This approach addresses the challenge where some unstable samples may not show clear abnormalities in initial cycles, potentially leading to premature classifications of stability. If the prediction lacks high credibility in determining stability or instability, the DF classifier continues monitoring post-fault until a credible instability result is obtained or a deadline is met. The detailed decision-making process is outlined in Algorithm 3.

Each output of the forest is a probability distribution, where the stable probability is denoted as p_{+1} and the unstable probability as p_{-1}. $\overline{p}^{l}$ represents the average value of the output probabilities of all trees at the l level. A stable judgment is only made when outputting samples for unknown samples in the last cycle, l_{max}. When the average stable output probability of the all-layer, $\overline{p}_{+1}$, exceeds α, the system is considered stable. This will minimize the possibility of judging unstable samples as stable ones to the greatest extent. In each cycle, unstable judgments are made for unknown samples to check whether the cycle exceeds the predefined threshold. $\overline{p}_{-1}$ represents the average instability probability threshold of all decision trees of the DF classifier, $\overline{p}_{-1}^{l-1}$ represents the average instability probability threshold of the decision tree of the previous layer of the DF classifier, and $\overline{p}_{-1}^{l}$ represents the average instability probability threshold of the decision tree of this layer of the DF classifier. For each evaluation cycle of data, when $b = M$, $\overline{p}_{-1} > \beta, \overline{p}_{-1}^{l-1} > \gamma, \overline{p}_{-1}^{l} > \delta$, and the system is considered unstable. If the above standards are not met, it is necessary to await the latest real-time data. These data will be incorporated into the subsequent evaluation cycle until reliable stability results are obtained or a deadline is met.

Algorithm 3. The proposed online assessment approach.

Two SSs are defined as "stable" and "unstable", labeled as "+1" and "−1" respectively. The output of each forest is a probability distribution (p_{+1}, p_{-1}).

Given M decision tree forests in level l, $a-1$ output and $b+1$ output. The output number of "+1" is a, and the output number of "−1" is b. $(a+b=M)$, $\alpha, \beta, \gamma, \delta$ are reliable thresholds. The result will only be considered reliable when the predicted probability is greater than the reliable threshold. The selection of reliable thresholds will be presented in Section 4.1.

While (the deadline is reached):

If: $a > M/2 \,\&\, \overline{p}_{+1} > \alpha$ →output "stable"

Else: output "unstable"

End while

While (the deadline is not reached):

If: $\begin{cases} b=M \\ \overline{p}_{-1} > \beta \\ \overline{p}_{-1}^{l-1} > \gamma \,\&\, \overline{p}_{-1}^{l} > \delta \end{cases}$ →output "unstable"

Else: no output

End while

The system state at t_p is 10 s after fault clearing in this context. If the post-fault assessment result is "stable," the SM of the new operation point is monitored by the DF regressor model. If the result is "unstable", system operators are immediately alerted. This timely alert allows emergency control measures to be implemented to protect the system from collapse.

4. Results and Discussion

In this section, an IEEE-39-bus test system model is used to validate the proposed unified approach for pre-fault and post-fault DSA, as depicted in Figure 7. This system encompasses 10 generators, 39 buses, 19 loads and 46 transmission lines, with PMUs assumed to be installed at each bus. The network standard topology serves as the foundation for the case study. To simulate the various operating conditions in the power system, all samples of the transient process are simulated and generated using the PSS/E platform. All generators use a fourth-order model. The generator connected to bus 39, Generator 10, is represented as an equivalent machine. Except for Generator 10, the excitation systems of all other generators adopt the IEEE DC1 excitation system model. All loads are modeled as constant impedance. All programs of the case study are compiled on a laptop with an Intel Core i7-9750H CPU working at 4.5 GHz and 8 GB RAM. An NVIDIA RTX-3050 GPU is configured to support efficient deep learning in Python 3.7.

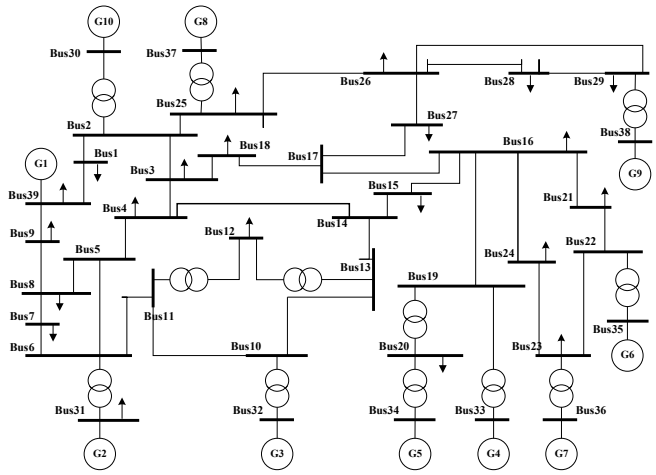

Figure 7. Network topology of the IEEE 39-bus system.

4.1. Parameter Sensitivity Analysis

Since the proposed unified DSA framework incorporates a time-adaptive strategy, the selection of the SM and the reliable threshold, $\alpha, \beta, \gamma, \delta$, significantly affects the assessment performance. It is necessary to determine the value of the SM and $\alpha, \beta, \gamma, \delta$ by parameter sensitivity analysis in fixed environments. The parameter sensitivity analysis is conducted by systematically varying the threshold values while keeping other parameters fixed, observing the changes in model performance to identify the optimal parameter settings. It is particularly important to emphasize that the parameter adjustment process was conducted on an independent validation set, which is completely separate from the test set, and there is no risk of data leakage. Furthermore, while the methodology presented in this paper is generalizable, the reliability thresholds vary across different systems to accommodate the unique physical stability margins and topological characteristics of each power system. Therefore, this paper conducts parameter sensitivity analyses for the IEEE 39-bus system and the 1648-bus system, with the details presented in Figures 8–10. As illustrated in Figure 8, the SM values of the stable samples in the 39-bus system mostly exceed 0.5, with the highest peak observed around 0.5. For the stable samples of the 1648-bus system, the SM values are predominantly greater than 0.9, with the peak value approximately centered around 0.9. Therefore, the SM values of the 39-bus system and the 1648-bus system are 0.5 and 0.9 respectively. The threshold α is used to determine whether the sample is stable. Therefore, when selecting the threshold, α, the accuracy of the stable sample judgments should be as high as possible. As shown in Figure 9, the threshold α values of the 39-bus system and the 1648-bus system are selected as 0.6 and 0.72. Figure 10 shows the values of β, γ, δ, which are used to determine whether the sample is unstable. When selecting the threshold β, γ, δ, the ART should be as small as possible while ensuring the accuracy of the judgment. In Figure 10, taking the threshold β as an example, increasing the threshold β does not yield significant performance improvements. On the contrary, it will increase ART to prolong the decision-making process. After comprehensive consideration, the values of β, γ, δ for the 39-bus system are 0.7, 0.6, and 0.6; the values of β, γ, δ for the 1648-bus system are 0.75, 0.69, and 0.69.

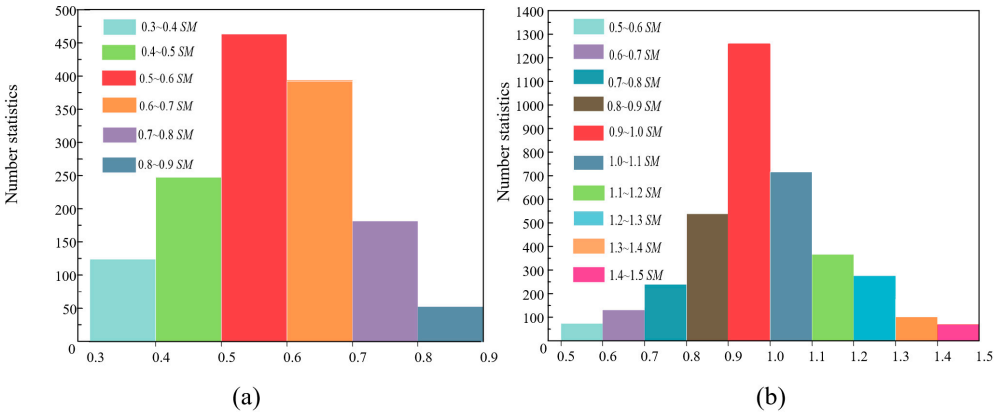

Figure 8. SM statistics of stable samples: (**a**) 39-bus system; (**b**) 1648-bus system.

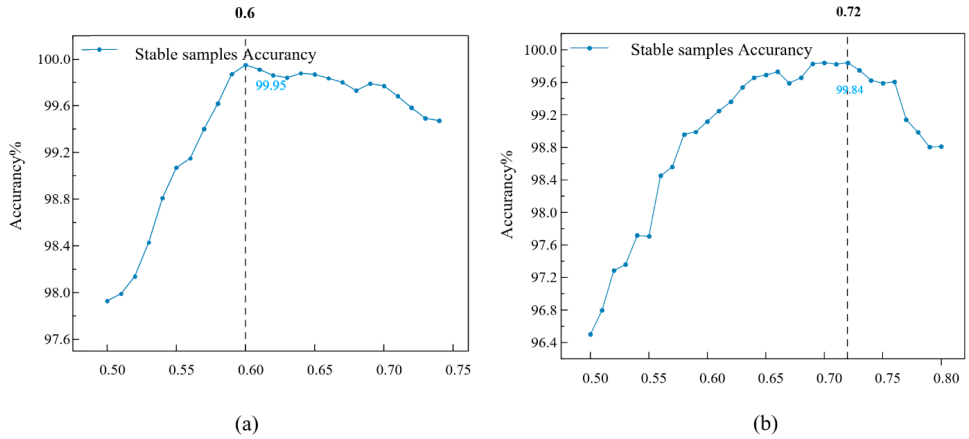

Figure 9. Sensitivity test for reliable threshold α: (**a**) 39-bus system; (**b**) 1648-bus system.

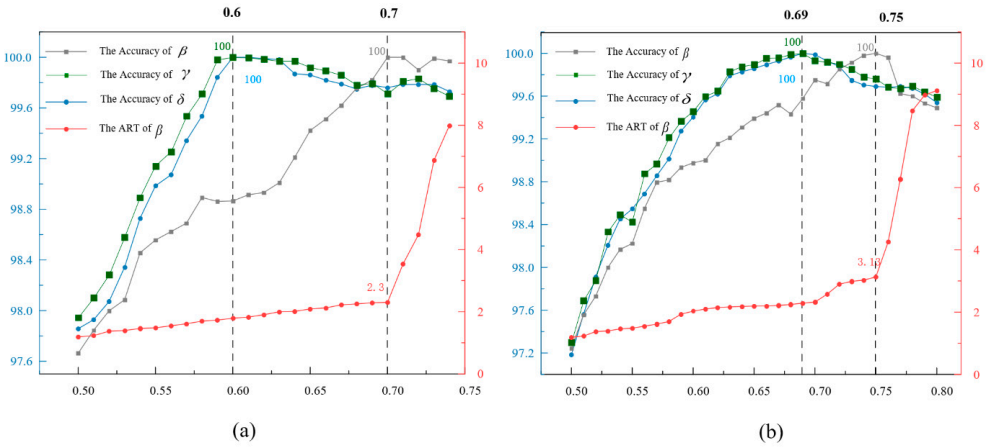

Figure 10. Sensitivity test for reliable threshold β,γ,δ: (**a**) 39-bus system; (**b**) 1648-bus system.

4.2. Performance of Feature Engineering

In this study, the raw features of the power system undergo processing through a MIC-based feature selection method and RSM-based feature extraction technology. Previous research, as referenced in [15,17], has demonstrated that artificially designed features exhibit a robust ability to reflect the behavior of a power system with small-scale data. Consequently, three distinct feature sets are taken into consideration for the comparative analysis. Feature set A is shown in Table 1.

Feature set A: The artificially designed features.
Feature set B: The features generated by the MIC and RSM.
Feature set C: The direct features sampled by PMUs.

Table 1. Feature set design.

Features	Description
1–2	Total active and reactive load of system
3–4	Total active and reactive power output
5	Max. generator rotor angle
6	Max. rotor angle variation
7	Average value of all generator rotor angles
8	Max. rotor acceleration
9	Min. rotor acceleration
10	Average value of all generator accelerations
11	Max. rotor acceleration variation
12	Rotor angle of generator with maximum acceleration
13	Rotor angle of generator with maximum kinetic energy
14	Total energy adjustment of the system
15	Max. generator rotor kinetic energy
16	Max. variation in rotor kinetic energy
17	Average value of rotor kinetic energy
18	Difference between max. and min. rotor angle
19	Difference between max. and min. rotor acceleration
20	Difference between max. and min. rotor kinetic energy
21	Max. generator rotor angular speed
22	Difference between max. and min. angular speed
23	Sum of generator rotor mechanical power

The amplitude of the bus voltage; the phase angle; the active and reactive power on the generator and the line; and the angle of the generator rotor are sampled as feature C. For feature set B, we select features with a correlation above 0.7 and remove those with a redundancy over 0.7, where n^{α} is set to 0.6 [26]. Subsequently, feature extraction is carried out using the RSM. As shown in Table 2, the models with different scanning window sizes achieve the best performance. This is because different window sizes enable the model to learn information at different time scales. In the RSM, three scanning windows with sizes of 5, 10, and 15 are employed. The above process results in the generation of 132 features for pre-fault and 64 features for post-fault DSA. The deadline of post-fault DSA is set to be 10 cycles. In total, 10 levels are trained for the DF classifier. The performance of the DF regressor and DF classifier is shown in Figures 11 and 12.

Table 2. Performance of deep forest model using different scan window sizes of RSM.

Window Size	DF Regressor	DF Classifier
	R^2	Accuracy
5	0.9846	98.35%
10	0.9869	98.76%
15	0.9856	98.64%
5,10,15	0.9889	98.95%

The residual squared error (R^2) is used to verify the performance of the proposed regression model, as shown in Equation (10):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}[y_i - \varphi(x_i)]^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2} \tag{10}$$

where n is the number of the test samples, y_i is the real value, $\varphi(x_i)$ is the prediction value, and $\overline{y}_i$ is the mean of y_i.

As illustrated in Figure 11, the performance of the DF regressor and DF classifier is superior for feature set B when compared with feature sets A and C. These results validate the effectiveness of the proposed feature processing method for DSA in power systems. Additionally, Figure 12 displays the training time of the proposed models. Notably, the training time of the DF classifier and DF regressor is the longest for feature set C and has the shortest training time in feature set B. This observation indicates the effectiveness of the proposed feature processing method for DSA in power systems.

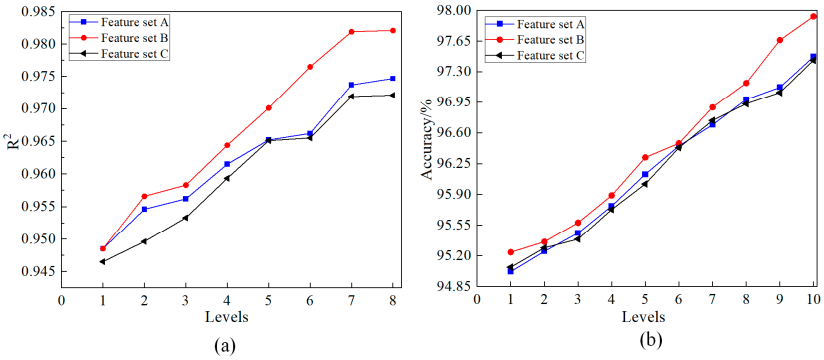

Figure 11. Performance of deep forest model using different input features: (**a**) DF regressor; (**b**) DF classifier.

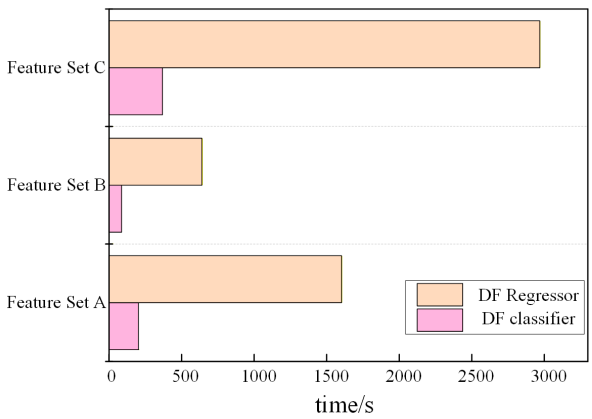

Figure 12. Training time of the proposed models using different feature sets.

4.3. Performance of the Proposed Pre-Fault DSA Model

In total, 1655 samples are utilized to train a DF regressor out of the 2069 total samples, leaving the remainder for testing. The performance of the DF regressor is evaluated using the R^2 and root mean square error ($RMSE$) metrics, as shown in Equation (11).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}[y_i - \varphi(x_i)]^2}{n}} \tag{11}$$

The performance of the DF regressor is assessed using the testing set, and the five-fold cross-validation results are presented in Table 3.

Table 3. Five-fold cross-validation results for DF regressor.

N-Folds	R^2	*RMSE*
1	0.9875	0.0124
2	0.9872	0.0118
3	0.9879	0.0119
4	0.9886	0.0125
5	0.9883	0.0114
Mean value	0.9879	0.0120

The results of the five-fold cross-validation indicate minimal variation across the folds, suggesting that the model is robust to data partitioning and possesses strong generalization capability. This also demonstrates that the DF regressor model exhibits high accuracy and stability in predicting the SM of the power system.

4.4. Performance of the Proposed Post-Fault DSA Model

A total of 8206 samples are selected for training, with 2052 reserved for testing, including 462 unstable and 1590 stable samples. The test results are summarized in Table 4.

Table 4. Performance of the stability assessment scheme.

Level	Stable Samples			Unstable Samples		
	Output	Error	Remained	Output	Error	Remained
1	0	0	1590	165	0	295
2	0	0	1590	137	0	163
3	0	0	1590	111	0	52
4	0	0	1590	21	0	31
5	0	0	1590	16	0	15
6	0	0	1590	3	0	12
7	0	0	1590	2	0	10
8	0	0	1590	2	0	8
9	0	0	1590	1	0	7
10	1583	7	0	14	0	0

The results illustrate that 90% of the unstable samples (a total of 413) are identified in the first three cycles, demonstrating that the proposed model is capable of rapidly detecting potential instability. Only seven unstable samples remain unrecognized until the last decision level, primarily because their characteristics were not clear at the initial stages, and they were correctly identified as more information became available. In the final cycle, the relatively conservative decision-making mechanism tends to predict instability, resulting in some stable samples being misclassified as unstable to avoid missing any unstable cases. Nevertheless, all unstable samples are identified successfully.

The average response time (ART) for the unstable samples, the average prediction accuracy (APA) for all decision cycles, and the prediction accuracy for all samples are employed to measure the entire performance of the proposed post-fault DSA model. The mathematical expressions of ART, accuracy and APA are shown in Equations (12)–(14).

$$ART = \sum_{i=1}^{I} [T_i \times C(T_i)] / \sum_{i=1}^{I} [C(T_i)] \tag{12}$$

$$Accuracy = (1 - (n_{\mathrm{fp}} + n_{\mathrm{fn}}) / n) \times 100\% \tag{13}$$

$$APA = \sum_{i=1}^{I} [A(T_i)] / I \tag{14}$$

where $C(T_i)$ is the total number of classified unstable samples in the current decision cycle; T_i is the decision time of the $i-th$ decision cycle; n_{fp} and n_{fn} are the number of stable samples assessed as unstable and the number of unstable samples assessed as stable, respectively; n is the total number of samples; $A(T_i)$ is the accuracy of current decision cycle; and I is the maximum allowable cycle.

To further verify the performance of the proposed model, as shown in Table 5, the five-fold cross-validation method was used for evaluation. The results indicate that the proposed DF classifier model has significant stability and reliability.

Table 5. Five-fold cross-validation results for DF classifier.

N-Folds	*ART*	*APA*	*Accuracy*
1	2.35 cycles	99.67%	99.54%
2	2.38 cycles	99.43%	99.49%
3	2.36 cycles	99.65%	99.59%
4	2.31 cycles	99.85%	99.68%
5	2.32 cycles	99.74%	99.64%
Mean value	2.34 cycles	99.67%	99.59%

4.5. Performance of the Unified DSA Framework

To evaluate the performance of the unified DSA framework, the load distribution is randomly varied between 80% and 120% of the base load level, with some loads ranging from 75% to 130%. The generator outputs are adjusted accordingly to maintain power balance. Based on these changes in load distribution and generation levels, 200 new OPs are generated. Power flow calculations are then performed, and any non-converging solutions are eliminated. As a result, 189 convergent OPs are retained. At each OP, a three-phase short-circuit fault is selected as the fault type. The total simulation duration is 10 s, with the fault occurring at 1 s into the normal operation of the system. The fault duration is randomly set between 0.05 s and 0.2 s. Three-phase-to-ground short-circuit faults are assumed to occur either at buses or at three locations (20%, 50%, and 80%) along each transmission line. This setup generates a total of 12,886 new samples. The number of training samples is 7833 (6066 stable samples and 1767 unstable samples), the number of validation samples is 2439 (1976 stable samples and 463 unstable samples), and the number of test samples is 2614 (2115 stable samples and 499 unstable samples). The unified DSA framework is constructed using the DF regressor and DF classifier. According to the parameter sensitivity analysis, 0.5 is set as the threshold of the SM. An operation point is deemed stable when the predicted SM exceeds 0.5. The online test results of the pre-fault DSA are presented in Table 6, where N_p represents the number of steady-state operation points, and N_{ps} and N_{pi} indicate the number of stable and unstable samples after the fault.

Table 6. Pre-fault DSA results for DF regressor for 39-bus system.

	N_p	N_{ps}	N_{pi}
SM > 0.5	107	1512	0
SM ≤ 0.5	82	603	499

As depicted in Table 6, the pre-fault DSA model identifies 107 operation points with a high SM value. When the system is subjected to anticipated faults in these operation points, the samples are directly considered stable. In practice, it is observed that all of these operation points indeed remain stable after the fault. The test results affirm the effectiveness of online pre-fault DSA.

The post-fault DSA model is applied to analyze an additional 82 operation points with SM values below 0.5 for online testing. The results are presented in Table 7. Table 7 indicates that 393 unstable samples are correctly classified within the first three cycles. The confusion matrix in Figure 13 shows that while 16 stable samples are mistakenly alerted, all unstable samples are predicted correctly. This can be attributed to the conservative nature of the time-adaptive decision logic defined in Algorithm 3, which effectively prevents misclassification of stable samples as unstable. This mechanism effectively eliminates the risk of false negatives.

Table 7. Post-fault DSA results for DF classifier for 39-bus system.

Level	Stable Samples			Unstable Samples		
	Output	Error	Remained	Output	Error	Remained
1	0	0	603	186	0	313
2	0	2	601	102	0	213
3	0	0	601	107	0	106
4	0	1	600	42	0	64
5	0	1	599	21	0	44
6	0	0	599	16	0	28
7	0	0	599	11	0	17
8	0	0	599	3	0	15
9	0	0	599	0	0	15
10	587	12	0	27	0	0

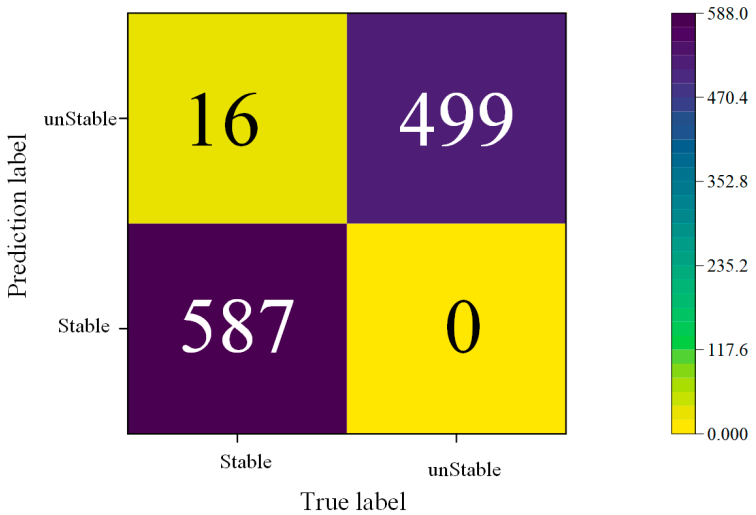

Figure 13. Confusion matrix for 39-bus system.

To test the performance of the proposed unified framework and the proposed model, we selected several common deep learning techniques for comparison, including CNN [14], LSTM [20], and GRU [12]. Additionally, we included representative deep learning models, including Transformer [15], GCN [16], and PA-AL [17]. These models were evaluated using both the proposed unified framework and the traditional post-fault assessment method for comparison. The parameters and implementation process are detailed

in the relevant sections of the literature. All models undergo five-fold cross-validation to eliminate the influence of random oversampling and adjust the model parameters through this process, thereby ensuring the stability of the model output results.

The results presented in Table 8 indicate that, under the same model conditions, although there is no significant difference in accuracy between the two evaluation frameworks, the proposed unified framework for pre-fault and post-fault has a noticeably shorter ART. This reduction in ART results from the pre-fault screening mechanism, which enables the system to bypass post-fault assessment for stable samples. The inference time results for the average inference time per cycle of each model show that, except for CNN, which is a fixed-window model, the average inference time per cycle for all models is shorter than the data evaluation period (16.66 ms at 60 Hz). This fully demonstrates the computational feasibility of the time-adaptive strategy in engineering applications. Additionally, when using the same evaluation framework, the proposed model outperforms all other models in both evaluation speed and accuracy, further validating the effectiveness of the proposed model in enhancing assessment efficiency.

Table 8. Comparison of different models for 39-bus system.

Model	Adopting Unified Framework				Not Adopting Unified Framework			
	ART	*APA*	*Accuracy*	Inference Time *	*ART*	*APA*	*Accuracy*	Inference Time *
Proposed	**2.66 cycles**	**98.89%**	**98.55%**	**1.52 ms**	**2.89 cycles**	**98.32%**	**98.11%**	**2.01 ms**
GRU	3.05 cycles	98.02%	97.86%	2.14 ms	3.96 cycles	97.87%	97.65%	2.95 ms
LSTM	3.27 cycles	97.32%	97.01%	2.45 ms	3.89 cycles	96.65%	96.81%	3.11 ms
CNN	10 cycles	97.83%	97.72%	17.12 ms	10 cycles	97.53%	97.53%	19.21 ms
Transformer	3.44 cycles	98.86%	98.65%	9.32 ms	4.27 cycles	98.22%	98.11%	10.12 ms
GCN	4.17 cycles	98.47%	98.56%	7.12 ms	5.21 cycles	98.15%	98.25%	7.94 ms
PA-AL	4.02 cycles	98.77%	98.54%	4.51 ms	4.69 cycles	98.53%	98.39%	5.39 ms

* The inference time in the table denotes the average computational time for the per-decision cycle. As CNN is a baseline model with fixed input, the inference time of CNN refers to the time required to process the entire fixed time window.

To verify the reliability of the proposed model, an ablation test is conducted under the unified online framework. In Table 9, by comparing the proposed model with MIC-DF and RSM-DF, it is proven that the performance of the integrated model proposed in this paper is not due to random variability. The simultaneous introduction of MIC and RSM enables the model to achieve a shorter judgment time and higher accuracy.

Table 9. Ablation test for 39-bus system.

Model	*ART*	*APA*	*Accuracy*
Proposed	**2.66 cycles**	**98.89%**	**98.55%**
MIC-DF	2.89 cycles	98.35%	98.26%
RSM-DF	2.96 cycles	97.98%	97.81%

4.6. Robustness Testing

During system operation, PMU data may be compromised by factors such as storage damage, network attacks, communication delays, or data loss. Additionally, the topological structure of the power system may change due to faults or maintenance activities. Therefore, the ability to evaluate performance under unknown topologies is a key indicator for the DSA model. In order to evaluate the robustness of the model under different conditions, four experimental scenarios were designed: missing data, noise interference, topology changes and various types of faults.

4.6.1. Testing for Missing Data

To examine the impact of missing data on the assessment accuracy of the DSA model, situations with different percentages of missing data are used in this test, and ρ is used to denote the percentage of missing data on the initial test set. In this test, ρ is set to 10%, 20%, and 30%. Table 10 shows the test of the DF regressor for the impact of missing data.

Table 10. Performance of DF regressor under missing data.

	R^2	*RMSE*
$\rho = 10\%$	0.9812	0.0156
$\rho = 20\%$	0.9731	0.0223
$\rho = 30\%$	0.9625	0.0389

According to Table 10, as the percentage of missing data increases, although the accuracy of the proposed DF regressor shows a downward trend, it maintains acceptable accuracy even when the missing data rate reaches 30%, demonstrating the model's structural tolerance to incomplete information.

Figure 14 shows the accuracy rate of the DF classifier at each level. Although the accuracy rates of different ρ values vary significantly at the beginning, as the cycle progresses and more information is obtained, the accuracy of different ρ values gradually converges. This can be attributed to the fact that the MIC and RSM enhance the diversity of the model by assigning different feature subsets to different trees, reducing the impact of missing specific features on the overall model performance. Even with a data missing rate of 30%, the model is still able to achieve an average accuracy of 97%.

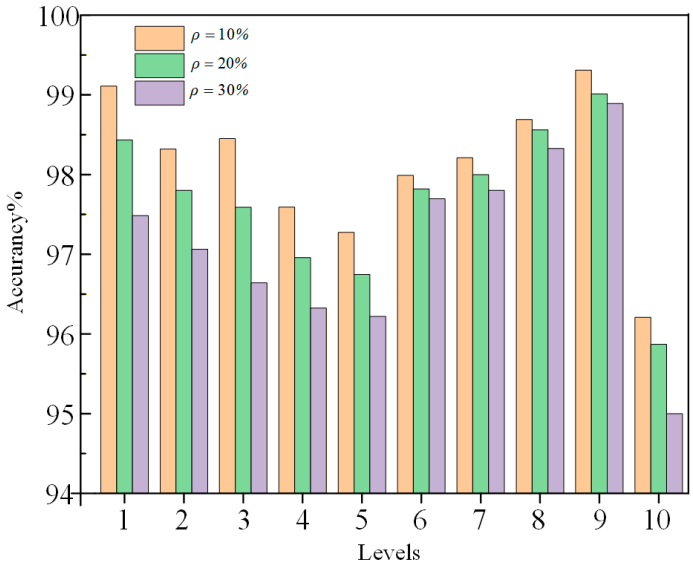

Figure 14. Performance of DF classifier considering the impact of missing data.

4.6.2. Testing for Noise Data

The measurement errors may be introduced in the sampling process. In fact, the PMU measurements can be expressed as

$$P_i(s) = P_i^{real}(s) + \varepsilon_{Pi}(s) \tag{15}$$

$$Q_i(s) = Q_i^{real}(s) + \varepsilon_{Qi}(s) \tag{16}$$

where $P_i^{real}(s)$ is the real value, and $\varepsilon_{Pi}(s)$ is the measurement error.

To test the impact of noise on the predictor, two scenarios are considered: (1) The random error $\varepsilon_{Pi}(s)$ has been added to the test dataset. (2) The random error has been added to both the training and test datasets. According to IEEE C37.118, "*Standard for Synchrophasors for Power Systems*" [29], the total vector errors of the standard PMUs should be less than 1%. Therefore, the random noise, $\varepsilon_{Pi}(s)$, changes in this range. The test results for the DF regressor are shown in Table 11.

Table 11. Performance of DF regressor under different scenarios.

	R^2	*RMSE*
Scenario 1	0.9012	0.0846
Scenario 2	0.9631	0.0377

As shown in Table 11, although the performance decreases in Scenario 1, with an R^2 of 0.9012, it remains within a functional range. Notably, when noise is introduced into the training set in Scenario 2, the performance is significantly restored, with R^2 reaching 0.9631, highlighting the adaptability of the model. Generally, effective measures must be taken to reduce the measurement noise. The adaptive Kalman filter is a recommended method for this purpose [30]. However, in most cases, the method itself is crucial for its robustness in handling missing data. To verify the robustness of this model, the experiment was conducted without using any noise filtering. Two scenarios are considered for the DF classifier. The test results are shown in Figure 15.

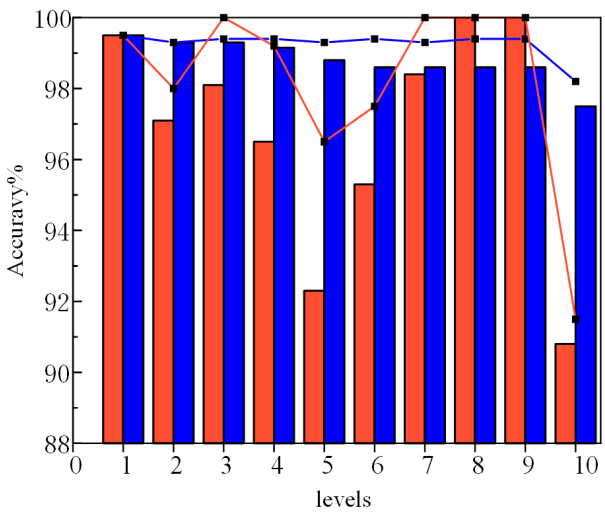

Figure 15. Performance of DF classifier considering the impact of noise.

In Figure 15, the histogram illustrates the test results of Scenario 1, and the curve represents the results of Scenario 2. The blue curve and histogram represent cumulative accuracy until the *i*-th cycle, and orange represents accuracy at the *i*-th cycle.

It can be observed that the DF classifier performs better in Scenario 2. This is because when noise is added to both the training and test sets, the model learns a broader range of features and patterns during training, enabling it to better handle the uncertainty present in real-world environments. The experimental results indicate that the classifier demonstrates satisfactory resilience to noise, maintaining stable performance with an accuracy of 98%, even without filtering. Therefore, it is highly suitable for use in DSA applications.

4.6.3. Testing for Topological Changes

This paper adopts the N-k topology structure, randomly disconnecting K branches to simulate line topological changes. Specifically, two datasets were generated: D_1 represents the scenario where branches 19–20 are disconnected, while D_2 represents the scenario where branches 2–3 and 16–21 are disconnected. The dataset generation process in this section is consistent with that in Section 4.2. An evaluation model was applied to test the new dataset, with results presented in Table 12, below.

Table 12. Model performance under topological changes.

Dataset	DF Regressor		DF Classifier	
	R^2	*RMSE*	*APA*	*Accuracy*
D_1	0.9721	0.0225	99.27%	99.12%
D_2	0.9689	0.0346	98.58%	98.31%

According to the results in Table 12, the proposed model demonstrates strong generalization ability when facing unknown topologies and maintains high performance even under scenarios involving power system topology changes.

4.6.4. Testing for Different Kinds of Faults

Since the faults considered in the model proposed in this paper are all three-phase faults, we refer to the data generation method described in Section 4.2 to consider other types of faults, including single-phase-to-ground faults, two-phase-to-ground faults, and two-phase faults. These additional fault scenarios form a dataset with dimension D_3, which will be used to test and evaluate the effectiveness of the proposed model, ensuring its robustness and general applicability under various fault conditions.

As shown in Table 13, the model demonstrates exceptional performance even under different fault types. This can be primarily attributed to the fact that three-phase faults represent the most severe transient disturbances in power systems, characterized by highly intense and complex dynamic behaviors. Since the model proposed in this paper has proven to effectively capture and handle the features under this worst-case scenario, it inherently possesses the generalization ability to manage milder disturbances, thereby maintaining a high level of robustness across different fault types.

Table 13. Model performance for different kinds of faults.

Dataset	DF Regressor		DF Classifier	
	R^2	*RMSE*	*APA*	*Accuracy*
D_3	0.9891	0.0121	99.56%	99.42%

4.7. Application to a Larger System

In order to comprehensively validate the unified DSA framework scheme, a practical 1648-bus system is selected as a larger test system, which includes 313 generators, 1220 load buses, 2294 transmission lines, and 182 shunts. The data generation method and tools used are the same as those used for the IEEE 39-bus test system. Ultimately, 372 operation points were solved, resulting in 24,930 samples. The number of training samples is 14,831 (11,762 stable samples and 3069 unstable samples), the number of validation samples is 4985 (3922 stable samples and 1063 unstable samples), and the number of test samples is 5114 (4192 stable samples and 922 unstable samples). The DF regressor and DF classifier are still used to build a unified DSA framework for testing. According to the parameter sensitivity analysis, 0.9 is set as the threshold of SM. The online test results of pre-fault

DSA are presented in Table 14, where the definitions of N_p, N_{ps} and N_{pi} are consistent with those of the IEEE 39-bus system.

Table 14. Pre-fault DSA results for DF regressor for 1648-bus system.

	N_p	N_{ps}	N_{pi}
SM > 0.9	234	3053	0
SM ≤ 0.9	138	1139	922

As demonstrated in Table 14, 234 operation points with high SM values remained stable even under fault conditions, further validating the applicability and effectiveness of the pre-fault DSA methodology in the 1648-bus system.

The post-fault DSA model is applied to analyze an additional 138 operation points with SM values below 0.9 for online testing. The results are presented in Table 15. The experimental results presented in Figure 16 demonstrate that, in the 1648-bus system, the time-adaptive strategy proposed in this paper remains effective, accurately identifying all unstable samples. This further validates its applicability in a large-scale power system.

Table 15. Post-fault DSA results for DF classifier for 1648-bus system.

Level	Stable Samples			Unstable Samples		
	Output	Error	Remained	Output	Error	Remained
1	0	0	1139	296	0	626
2	0	0	1139	212	0	412
3	0	3	1136	157	0	258
4	0	0	1136	101	0	157
5	0	4	1132	75	0	86
6	0	0	1132	36	0	50
7	0	2	1130	15	0	37
8	0	1	1129	9	0	29
9	0	0	1129	6	0	23
10	1104	25	0	48	0	0

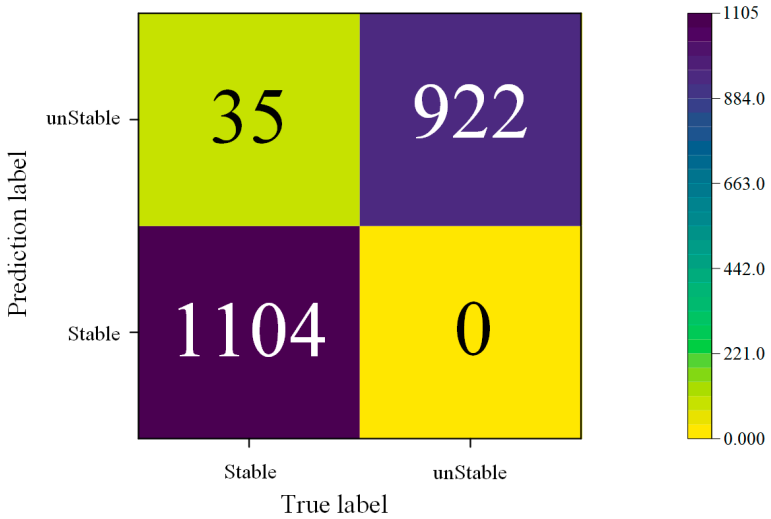

Figure 16. Confusion matrix for 1648-bus system.

Table 16 validates the scalability of the proposed framework, demonstrating that it maintains good effectiveness even in the large-scale 1648-bus system. Under consistent evaluation conditions, the proposed model still outperforms other benchmark models.

Compared with the 39-bus system, although the performance of ART and accuracy slightly declined, the model still maintained the ART of 3.13 cycles (0.0521) and 98.34% accuracy, and the inference time of each model remains lower than the PMU evaluation period, which is sufficient for online applications. This can be attributed to the multi-layer integrated structure of DF, which can enhance the computing efficiency and performance of large-scale systems.

Table 16. Comparison of different models for 1648-bus system.

Model	Adopting Unified Framework				Not Adopting Unified Framework			
	ART	*APA*	*Accuracy*	Inference Time *	*ART*	*APA*	*Accuracy*	Inference Time *
Proposed	**3.13 cycles**	**97.46%**	**98.34%**	**1.89 ms**	**3.37 cycles**	**97.01%**	**98.05%**	**2.65 ms**
GRU	4.05 cycles	97.11%	96.64%	2.97 ms	4.86 cycles	96.46%	96.22%	3.79 ms
LSTM	5.11 cycles	96.32%	96.47%	3.11 ms	5.73 cycles	95.89%	96.05%	3.94 ms
CNN	10 cycles	96.88%	96.88%	19.56 ms	10 cycles	96.21%	96.21%	21.21 ms
Transformer	4.87 cycles	96.97%	96.78%	11.49 ms	5.54 cycles	96.41%	96.05%	12.52 ms
GCN	5.01 cycles	96.46%	96.39%	8.41 ms	5.97 cycles	95.97%	96.24%	9.24 ms
PA-AL	4.27 cycles	97.41%	97.68%	5.11 ms	5.22 cycles	96.89%	96.92%	5.88 ms

* The inference time in the table denotes the average computational time for the per-decision cycle. As CNN is a baseline model with fixed input, the inference time of CNN refers to the time required to process the entire fixed time window.

5. Conclusions

In order to enhance the reliability of data-driven DSA in practical power systems, this paper proposes a unified online DSA framework that seamlessly integrates both pre-fault and post-fault assessment processes. When the system encounters a fault, the framework predicts the stability metric (SM) and instability metric (SS) using DF regressors and classifiers. Through simulation analysis and comparative experiments, the following conclusions are drawn:

(1) The unified framework proposed in this paper introduces the concept of safety margin division. A key feature of this framework is its ability to construct both pre-fault and post-fault DSA, focusing on the global timeline of disturbance occurrence. The method considers scenarios with predicted SM exceeding the threshold as stable and applies post-fault DSA only to scenarios with SM below the threshold to determine SS. Numerical tests on the IEEE 39-bus system and a practical 1648-bus system demonstrate that the proposed framework successfully predicts stable scenarios through pre-fault DSA and unstable scenarios through post-fault DSA, achieving notably short response times in the online stage.

(2) The proposed assessment method employs feature engineering, utilizing the MIC and RSM to identify key variables. The MIC ensures that each decision tree focuses on features highly correlated with system stability, while the RSM allows different trees to learn combinations of feature subsets. This synergy enables the model to effectively capture the dynamic behavior of complex systems. Robustness testing results show that the method maintains high performance and strong robustness even when facing missing data, noise interference, or large-scale datasets.

(3) In addition, this paper introduces a time-adaptive scheme that eliminates the reliance on fixed time length judgments and allows for dynamic adjustment of assessment thresholds based on the specific characteristics of different systems. This ensures high prediction accuracy and fast response times.

Author Contributions: Conceptualization, X.L., R.S. and Q.Z.; methodology, R.S. and Y.Z.; software, R.S.; validation, X.L., and J.L.; formal analysis, C.W.; investigation, P.G.; resources, Q.Z.; data curation, X.L.; writing—original draft preparation, R.S. and Y.Z.; writing—review and editing, R.S.;

visualization, J.L.; supervision, C.W.; project administration, P.G.; funding acquisition, Q.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China, grant number 52107107; the Key Science and Technology Projects of Yunnan Province, grant number 202503AA080001; and the Doctoral Research Startup Fund of Suzhou University, grant number2023BSK055.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding authors.

Acknowledgments: We express sincere gratitude to all those who have supported this research.

Conflicts of Interest: Author Qiao Zhao was employed by the company Power China Guiyang Engineering Corporation Limited. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Sarajcev, P. Machine learning in power system dynamic security assessment. *Energies* **2022**, *15*, 3962.
2. Fathollahi, A. Machine learning and artificial intelligence techniques in smart grids stability analysis: A review. *Energies* **2025**, *18*, 3431.
3. Yang, C.; Li, H.; Zeng, W.; Fan, J.; Ren, Z. Fault detection and classification of power lines based on Bayes–LSTM–Attention. *Energies* **2025**, *18*, 6483.
4. Xu, Y.; Dong, Z.Y.; Xu, Z.; Meng, K.; Wong, K.P. An intelligent dynamic security assessment framework for power systems with wind power. *IEEE Trans. Ind. Informat.* **2012**, *8*, 995–1003.
5. Li, X.; Yang, Z.; Guo, P.; Cheng, J. An intelligent transient stability assessment framework with continual learning ability. *IEEE Trans. Ind. Informat.* **2021**, *17*, 8131–8141.
6. La Scala, M.; Sbrizzai, R.; Torelli, F.; Scarpellini, P. A tracking time domain simulator for real-time transient stability analysis. *IEEE Trans. Power Syst.* **1998**, *13*, 992–998.
7. Xue, Y.; Van Custem, T.; Ribbens-Pavella, M. Extended equal area criterion justifications, generalizations, applications. *IEEE Trans. Power Syst.* **1989**, *4*, 44–52.
8. Vittal, V.; Zhou, E.Z.; Hwang, C.; Fouad, A.-A. Derivation of stability limits using analytical sensitivity of the transient energy margin. *IEEE Trans. Power Syst.* **1989**, *4*, 1363–1372.
9. Zhang, Y.; Xu, Y.; Dong, Z.Y. Robust ensemble data analytics for incomplete PMU measurements-based power system stability assessment. *IEEE Trans. Power Syst.* **2018**, *33*, 1124–1126.
10. Liu, S.; Liu, L.; Fan, Y.; Zhang, L.; Huang, Y.; Zhang, T.; Cheng, J.; Wang, L.; Zhang, M.; Shi, R.; et al. An integrated scheme for online dynamic security assessment based on partial mutual information and iterated random forest. *IEEE Trans. Smart Grid* **2020**, *11*, 3606–3619.
11. Zhu, L.; Lu, C.; Luo, Y. Time series data-driven batch assessment of power system short-term voltage security. *IEEE Trans. Ind. Informat.* **2020**, *16*, 7306–7317.
12. Ren, C.; Xu, Y.; Zhang, R. An interpretable deep learning method for power system transient stability assessment via tree regularization. *IEEE Trans. Power Syst.* **2022**, *37*, 3359–3369.
13. Liu, S.; Liu, S.; Li, X.; Gu, Y.; Li, Z.; Yang, C.; Zhang, Y.; Hu, J.; Tan, R.; Liu, C. Neyman-Pearson Umbrella Algorithm-Based Static Voltage Stability Assessment With Misclassification Restriction: An Integrated Data-Driven Scheme. *IEEE Trans. Ind. Informat.* **2023**, *19*, 11391–11402.
14. Zhao, S.L.; Kai, Z. Transient stability assessment of power system based on improved one-dimensional convolutional neural network. *Power Syst. Technol.* **2021**, *45*, 2945–2957.
15. Wang, H.; Gao, F.; Chen, Q.; Bu, S.; Lei, C. Instability Pattern-Guided Model Updating Method for Data-Driven Transient Stability Assessment. *IEEE Trans. Power Syst.* **2025**, *40*, 1214–1227.
16. Xia, S.; Liu, Y.; Zhang, T. GCN-LSTM Based Transient Angle Stability Assessment Method for Future Power Systems Considering Spatial-Temporal Disturbance Response Characteristics. *Protect. Control Mod. Power Syst.* **2024**, *9*, 108–121.

17. Shen, C.; Zuo, K.; Sun, M. Physics-Augmented Auxiliary Learning for Power System Transient Stability Assessment. *IEEE Trans. Ind. Informat.* **2025**, *21*, 6811–6822.
18. Fang, J.; Li, M.; Zhang, X. A data-driven method for online transient stability monitoring with vision-transformer networks. *Int. J. Electr. Power Energy Syst.* **2023**, *149*, 109020.
19. Shao, Z.; Wang, Q.; Cao, Y.; Cai, D.; You, Y.; Lu, R. A Novel Data-Driven LSTM-SAF Model for Power Systems Transient Stability Assessment. *IEEE Trans. Ind. Informat.* **2024**, *20*, 9083–9097.
20. Azhar, I.F.; Putranto, L.M.; Irnawan, R. Development of PMU-based transient stability detection methods using CNN-LSTM considering time series data measurement. *Energies* **2022**, *15*, 8241.
21. Ren, C.; Yuan, H.; Li, Q.; Zhang, R.; Xu, Y. Pre-Fault Dynamic Security Assessment of Power Systems for Multiple Different Faults via Multi-Label Learning. *IEEE Trans. Power Syst.* **2023**, *38*, 5501–5511.
22. Lu, G.; Bu, S. Online Power System Dynamic Security Assessment: A GNN–FNO Approach Learning From Multi-source Spatial–Temporal Data. *IEEE Trans. Ind. Informat.* **2025**, *21*, 7598–7608.
23. Sobbouhi, A.R.; Vahedi, A. Transient stability prediction of power system; a review on methods, classification and considerations. *Electr. Power Syst. Res.* **2021**, *190*, 106853.
24. Moulin, L.S.; Liu, S.; Zhang, X. Support vector machines for transient stability analysis of large-scale power systems. *IEEE Trans. Power Syst.* **2004**, *19*, 818–825.
25. Zhou, Z.; Feng, J. Deep forest. *Nat. Sci. Rev.* **2019**, *6*, 74–86.
26. Reshef, D.N.; Reshef, Y.A.; Finucane, H.K.; Grossman, S.R.; McVean, G.; Turnbaugh, P.J.; Lander, E.S.; Mitzenmacher, M.; Sabeti, P.C. Detecting novel associations in large data sets. *Science* **2011**, *334*, 1518–1524.
27. Zhu, L.; Hill, D.J.; Lu, C. Hierarchical deep learning machine for power system online transient stability prediction. *IEEE Trans. Power Syst.* **2019**, *35*, 2399–2411.
28. Zhang, R.; Xu, Y.; Dong, Z.Y.; Wong, K.P. Post-disturbance transient stability assessment of power systems by a self-adaptive intelligent system. IET Gener. *Transm. Distrib.* **2015**, *9*, 296–305.
29. *C37.118-2005*; IEEE Standard for Synchrophasors for Power Systems. IEEE: Piscataway, NJ, USA, 2005.
30. Zhang, G.; Gao, W.; Li, Y.; Guo, X.; Hu, P.; Zhu, J. Detection of false data injection attacks in a smart grid based on WLS and an adaptive interpolation extended Kalman filter. *Energies* **2023**, *16*, 7203.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.